TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

5-Anchor Subitizing, Regrouping & Base-10 Manipulatives

SERIES 1 • CARD 01 / 12 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING

CARD 01 / 12 • BASE-10

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

5-ANCHOR SUBITIZING & BRIDGING TO 10

Ten-frames visually organize numbers into 5-anchors:

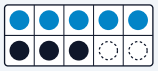
5-Anchor Formula: 5 + n • Ten's Comp: 10 - n

Seeing 8 as 5 on top and 3 below anchors addition, turning counting into instant visual recognition!

CALIBRATED 14.00 MM SUBITIZING GRID

- Counter Diameter: 14.00 mm (2U) with two-tone face
- Stacking Datum: 1.85mm base plug & stacking socket
- Ten-Frame Grid: 2 × 5 array with 14.00 mm wells

5-ANCHOR SUBITIZING: 5 + 3 = 8 (BRIDGE 10)



INSTANT SUBITIZING

Top Row: 5 Blue (anchor)

Bottom: 3 Red (8 total)

2 Empty Slots:

10 - 8 = 2 Ten's Comp

Bridge: 8 + 5 = 13

2. KIT PARTS INVENTORY (56 PIECES)

COMPONENT	QTY	SPEC	COLOR
10-Frame Trays	2x	97.4×38.6mm	PLA Gray
Counter Blocks A	26x	19.0 mm Tile	PLA White
Counter Blocks B	26x	19.0 mm Tile	PLA Tan
Regrouping Cubes	2x	19.0 mm Cube	Muted Red
Twin Stage Lid	1x	Twin 10-Frame	PLA White
3x1 Storage Bin	1x	49.0 mm	PLA White

TOTAL: 52 Counters + 2 Carry Cubes + 2 Trays + Box + Lid = 56 Parts

SLANT 3D PRODUCTION PALETTE

- PLA White: Counter Set A, Storage Box Body & Stage Lid.
- PLA Tan: Counter Set B for second operand.
- PLA Gray: Dual 2x5 activity trays for neutral ground.
- PLA Matte Muted Red: Regrouping carry cubes (+1 / +10).

3. PACK-UP & SUPPORT

✓ **10-SECOND PACK-UP**

Populate both 10-frames with 10 counters each and lay carry cubes into center trough. Empty wells instantly spot missing pieces. Twin stage lid snaps flush.

✓ **NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ **HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Base-10 positional notation decomposes any integer into powers of 10 with digits 0 through 9.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: MAKE-A-TEN BRIDGE & 5-ANCHOR SUBITIZING

Step-by-Step Hands-On Tutorial: 8 + 5 = 13 via Bridging to 10

SERIES 1 • CARD 01 OF 12 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING

CARD 01 / 12 • BASE-10

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/ten-frames/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 1.OA.C.6 (Make Ten)
- CCSS 2.NBT.B.5 (Place Value)
- TEKS 111.3.b.3.C (Compose/Deco...)

Grades K–3 Subitizing & Place Value

GUIDING INQUIRY

How many empty spots are left in Frame A before you start moving counters from Frame B?

1 STEP 1: POPULATE 8 ON TEN-FRAME A

Place 8 blue counters in Ten-Frame A.

Fill the top row (5 blue) and bottom row (3 blue).

Instant Subitizing: 5 + 3 = 8 (zero counting needed!).

Notice: exactly 2 slots remain empty to

2 STEP 2: POPULATE 5 ON TEN-FRAME B

Place 5 red counters in Ten-Frame B (fills top row of 5).

We want to calculate 8 + 5 without counting by ones.

Frame A needs 2 counters to bridge to 10.

3-4 STEPS 3 & 4: BRIDGE TO 10 & REGROUP

3. BRIDGE TO 10: Move 2 red counters from B to A. Frame A is now 1 Ten (10)!
4. READ REMAINING ONES: Frame B has 3 red counters left. Total: 10 + 3 = 13!

Keystrokes: Ten-Frame Icon ([▦] & [PV]) 8 [ENTER] 2 [+] 5 [+] [▦] → 15.

5 CORE PRINCIPLE: ADDITION IS BRIDGING, NOT COUNTING

Counting on fingers by ones is slow and error-prone. The 10-frame proves that addition is decomposing numbers into friendly anchors: 8 + 5 = 8 + (2 + 3) = (8 + 2) + 3 = 10 + 3 = 13! Once students see the 2 empty spaces in the 8-frame, they instantly know to borrow 2 from the 5!

★ EXTENSION INVESTIGATION: BRIDGE TO 10: 7 + 6

Put 7 in Frame A (3 empty). Put 6 in Frame B. Move 3 to make a ten. How many are left in B? 10 + 3 = 13! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL

stackcalc.io/parts

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH

Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC

Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY

Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING

Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5

Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC

[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS

Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS

Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING

Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1

Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION

Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS

Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED

Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS

Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4

Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:

Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:

Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:

Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:

Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

BASE-10: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:

All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:

Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:

Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:

2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:

100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:

Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 56 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California